

PH4102: Introductory Astrophysics

Instructor: Sudip K Garain

Contact information

Office: 221B AAC Building

Email: sgarain@iiserkol.ac.in

Class timings

- **Monday : 10:45 AM – 11:40 AM**
- **Tuesday : 9:50 AM – 10:45 AM**
- **Thursday : 8 PM – 8:55 AM (tutorial), 8:55 AM – 9:50 AM**

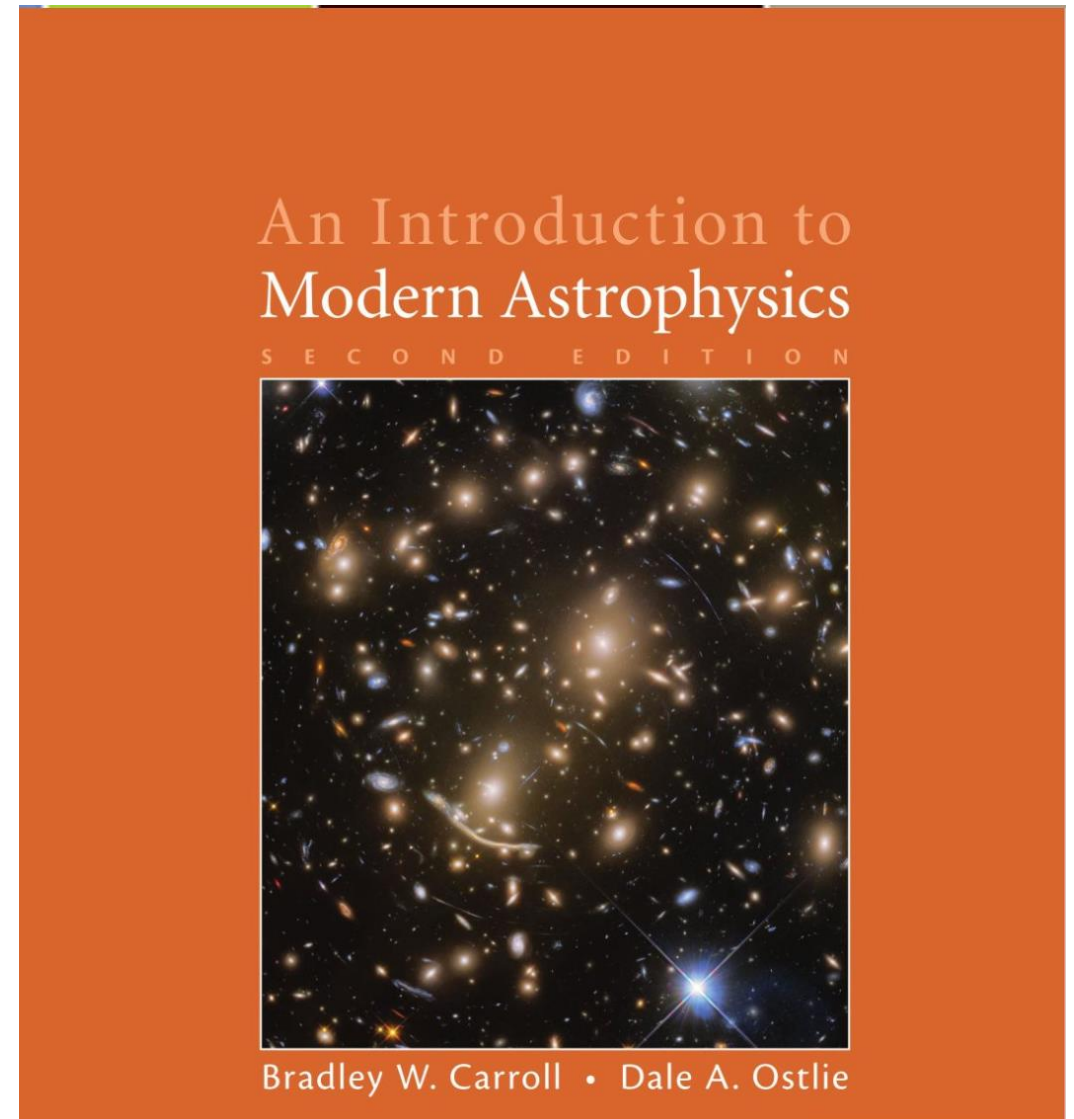
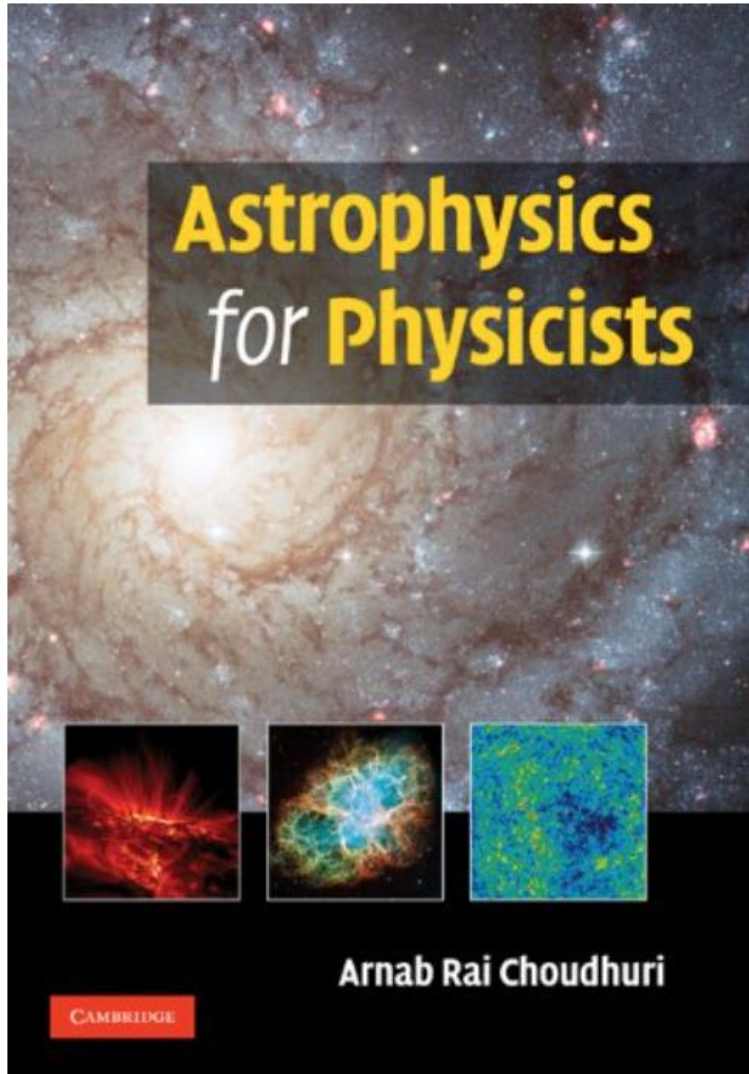
Tentative Grading Scheme

- Final Exam : 50
- Mid-Sem : 30
- Class tests : 5 + 5 (both based on homework problems)
- Quiz : 5 (6-7 quizzes each of 10 mark and best 5 considered!
Finally scaled to 5)
 - On Thursdays (20th Aug, 3rd Sep, 24th Sep, 8th Oct, 29th Oct, 12th Nov, 19th Nov)
- Attendance : 5 (more than 80% preferable)

Broad overview of this course

- Basic astronomy (measurement units, coordinate system, observation, celestial mechanics etc.)
- Radiation, radiative transfer equation, solution, physical systems
- Stars – structure, equations, nucleosynthesis, evolution
- Stars – collapse, degeneracy pressure, compact objects
- Star formation, ISM, molecular clouds, star clusters
- Plasma astrophysics
- Cosmology

Suggested books:



Additional books:

- **M S Longair, High Energy Astrophysics**
- **Zeilik & Gregory, Introductory Astronomy & Astrophysics**
- **Dan Maoz, Astrophysics in a Nutshell**